



Big Databases : NoSQL





Before all :

let's introduce ourselves

- Me :
 - Training
 - Work experience
- You :
 - Initial training
 - Work/internship experience
 - Level of knowledge in databases, NoSQL, Big data, docker
 - Expectations for this course



Objectives

- 1 : Conceptual understanding of NoSQL databases
- 2 : Proficiency in setting up experimental environment for big databases
- 3 : Master Core NoSQL Databases
- 4 : NoSQL databases integration and choosing the right tool
- 5 : Implement Real-World Data Solutions



Objective 1 : Conceptual understanding of NoSQL databases

- The evolution of data
- Definition of big data
- Traditional databases (RDBMS) limitations
- Introduction to NoSQL Databases



Objective 2 : Proficiency in setting up experimental environment for big databases

- Introduction to containerization
- Docker for Database Environments



Objective 3 : Master Core NoSQL Databases

- Key-Value Stores
- Document Stores
- Column-Family Stores
- Graph Databases



Objective 4 : NoSQL databases integration and choosing the right tool

- When to use each type of NoSQL database?
- Data ingestion strategies (ETL/ELT)
- Introduction to Spark & NoSQL databases



Objective 5 : Implement Real-World Data Solutions

- Presentation and implementation of a global use case



Evaluation methods





Evaluations

- A series of application exercises to be completed after lessons no. 3 and 7
- Participation grade
- Final project : last day



My rules

- => Avoid Chatter : talk about the course, ask questions!
- => No wrong question/answer
- => Give your thoughts, share your knowledge
- => **DO NOT USE CHATGPT** or any other AI => try by yourself first : penalties for those who use it



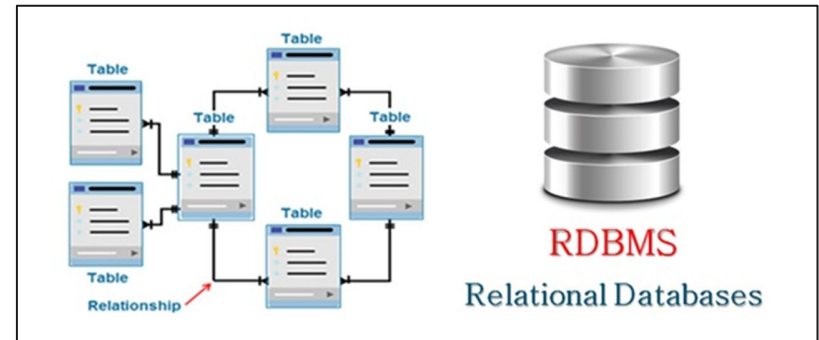
Part 1 :

Big Data Concepts and
Challenges

The evolution of data: from structured to big data

Structured data

- Organized in tables (rows and columns)
- Stored in relational databases
- Data managed with SQL in relational DBMSs
- Examples of DBMS :
 - MySQL
 - PostgreSQL
 - Oracle





The evolution of data: from structured to big data

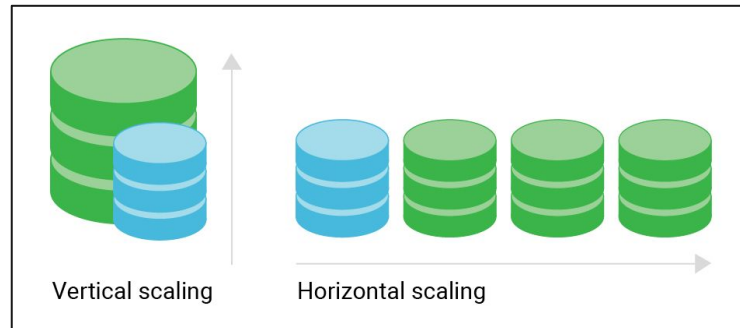
Structured data

- Advantages of relational DBMS :
 - Enforces strong schema (data types, constraints)
 - Types : String, Integer, Boolean, Date, ...
 - Relations : PK, Unique, Non NULL, FK, ...
 - Easy to query with SQL
 - ACID Transactions :
 - **Atomicity** : A transaction must be fully completed or not executed at all; **All or Nothing**
 - **Consistency** : Transactions must bring the DB from one valid state to another; **Always Valid State**
 - **Isolation** : Multiple transactions can occur simultaneously without interfering with each other; **No Cross-Talk**
 - **Durability** : Once a transaction is committed, it will persist even if there's a crash; **Persistence Guaranteed**

The evolution of data: from structured to big data

Structured data

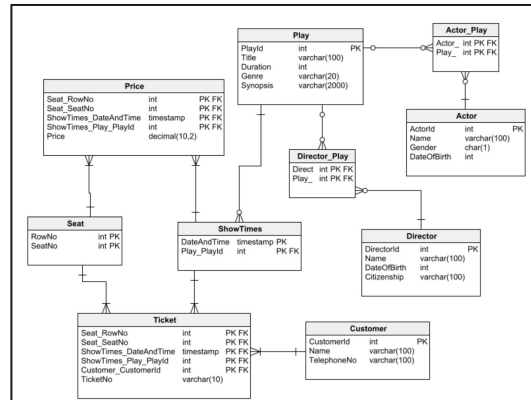
- Limitations :
 - **Scalability** : only optimized for vertical scalability (adding more power to a single server) and not for horizontal scalability (adding more servers)
=> No possibility to use “sharding” (splitting data across servers)



The evolution of data: from structured to big data

Structured data

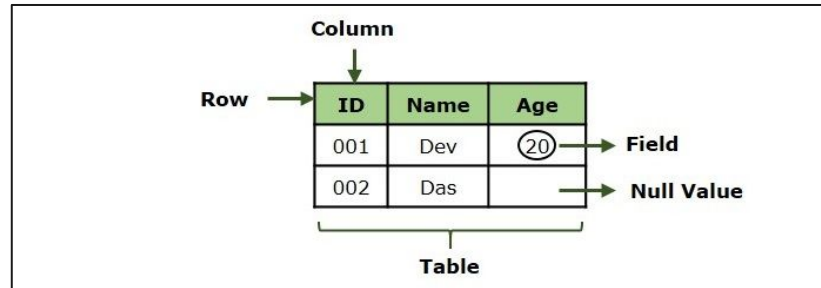
- Limitations :
 - **Schema rigidity** : Adding or modifying columns in production often requires downtime or complex migration scripts
 - Constraints : type, relations ...



The evolution of data: from structured to big data

Structured data

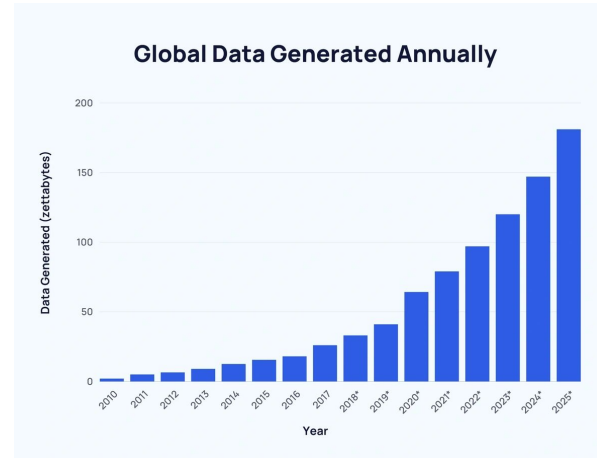
- Limitations :
 - **Inflexibility** : Not optimized for semi-structured, unstructured, hierarchical or high-velocity (real-time) data
 - Images, Audio, Documents, Videos, JSON...



The evolution of data: from structured to big data

Big Data Era

- Explosion of data from:
 - Social media



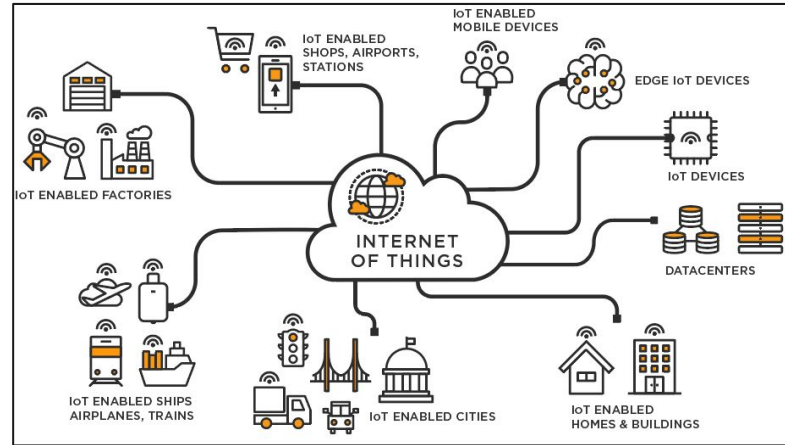
Multiples of bytes				v · d · e
SI decimal prefixes		Binary	IEC binary prefixes	
Name (Symbol)	Value	usage	Name (Symbol)	Value
kilobyte (kB)	10 ³	2 ¹⁰	kibibyte (KiB)	2 ¹⁰
megabyte (MB)	10 ⁶	2 ²⁰	mebibyte (MiB)	2 ²⁰
gigabyte (GB)	10 ⁹	2 ³⁰	gibibyte (GiB)	2 ³⁰
terabyte (TB)	10 ¹²	2 ⁴⁰	tebibyte (TiB)	2 ⁴⁰
petabyte (PB)	10 ¹⁵	2 ⁵⁰	pebibyte (PiB)	2 ⁵⁰
exabyte (EB)	10 ¹⁸	2 ⁶⁰	exbibyte (EiB)	2 ⁶⁰
zettabyte (ZB)	10 ²¹	2 ⁷⁰	zebibyte (ZiB)	2 ⁷⁰
yottabyte (YB)	10 ²⁴	2 ⁸⁰	yobibyte (YiB)	2 ⁸⁰

See also: Multiples of bits · Orders of magnitude of data

The evolution of data: from structured to big data

Big Data Era

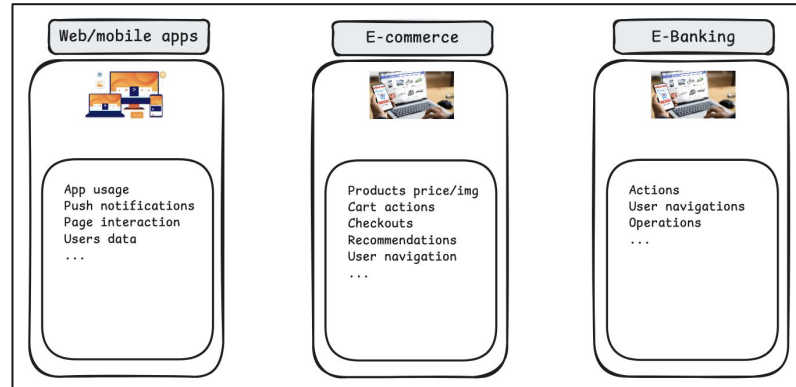
- Explosion of data from:
 - IoT



The evolution of data: from structured to big data

Big Data Era

- Explosion of data from:
 - Logs and clickstreams



Require new models for storage & processing : real time, various formats, ...



The evolution of data: from structured to big data

Characteristics of Big Data

- **Volume** – Massive data size (terabytes to petabytes)
- **Velocity** – Data generated in real-time
- **Variety** – Structured, semi-structured, unstructured
- **Veracity** – Quality & uncertainty of data
- **Value** – Turning data into insights

Transition to NoSQL

Types of Modern Data

Structured



Semi-Structured



JSON

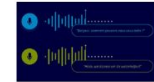


XML



CSV

Unstructured





Transition to NoSQL

Concept of NoSQL

- NoSQL ⇔ Not Only SQL
- Databases that allows you to manage the limitations of RDBMS :
 - **Flexible schema** : can store different data types
 - **Horizontal scalability** : Easily scale out by adding more servers
 - **High performance for specific workloads** : massive data, real time
 - **Availability and partition tolerance** : distribute data over servers



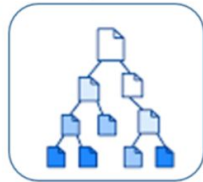
Part 2 :

Introduction to NoSQL Databases

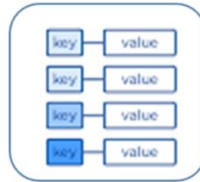
NoSQL Databases

Classification

Store semi-structured data as documents (typically JSON, XML....)
Use cases : user profiles, product catalogs
Ex. : MongoDB, DocumentDB

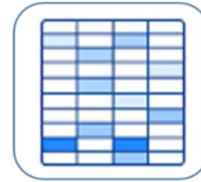


Document Store



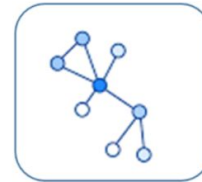
Key-Value Store

Simple data stores using unique keys to access values
Use cases : Caching, shopping carts
Ex. : Redis, DynamoDB



Wide-Column Store

Store data in column families that can be distributed across nodes
Use cases : event logging, IoT applications
Ex. : Cassandra, HBase



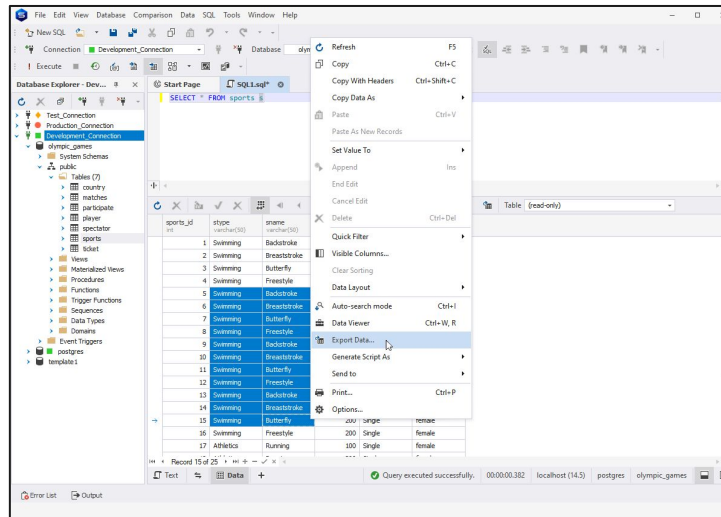
Graph Store

Specialized for highly connected data with explicit relationship modeling
Use cases : Social networks, recommendation engines
Ex. : Neo4j, JanusGraph

NoSQL Databases

Classification

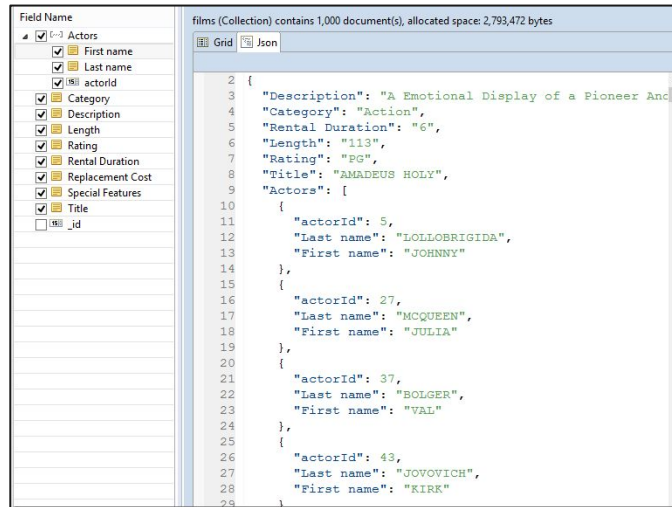
- Relational databases :



NoSQL Databases

Classification

- Document databases :



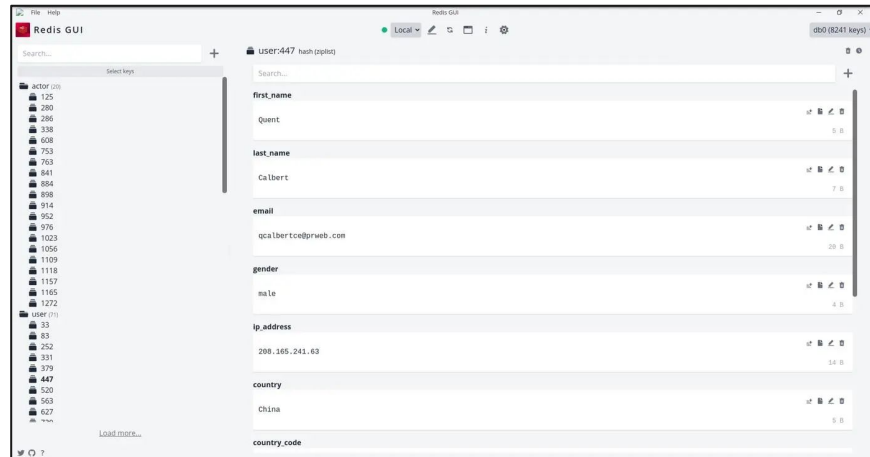
The screenshot displays a NoSQL database interface. On the left, a 'Field Name' list shows various fields with checkboxes, including 'First name', 'Last name', 'actorId', 'Category', 'Description', 'Length', 'Rating', 'Rental Duration', 'Replacement Cost', 'Special Features', 'Title', and '_id'. The main area shows a document in a collection named 'films'. The document is displayed in a grid view and contains the following JSON structure:

```
{
  "Description": "A Emotional Display of a Pioneer And",
  "Category": "Action",
  "Rental Duration": "6",
  "Length": "113",
  "Rating": "PG",
  "Title": "AMADEUS HOLY",
  "Actors": [
    {
      "actorId": 5,
      "Last name": "LOLLOBRIGIDA",
      "First name": "JOHNNY"
    },
    {
      "actorId": 27,
      "Last name": "MCQUEEN",
      "First name": "JULIA"
    },
    {
      "actorId": 37,
      "Last name": "BOLGER",
      "First name": "VAL"
    },
    {
      "actorId": 43,
      "Last name": "JOVOVICH",
      "First name": "KIRK"
    }
  ]
}
```

NoSQL Databases

Classification

- Key-value databases :



NoSQL Databases

Classification

- Column-family databases :

Each cell has multiple versions, typically represented by the timestamp of when they were inserted into the table

Timestamp1 Timestamp2

Row Key	Column Family - Personal		Column Family - Office		Address
	Name	Residence Phone	Phone		
00001	John	415-111-1234	415-212-5544		1021 Market St
00002	Paul	408-432-9922	415-212-5544		1021 Market St
00003	Ron	415-993-2124	415-212-5544		1021 Market St
00004	Rob	815-243-9988	408-998-4322		4455 Bird Ave
00005	Carly	206-221-9123	408-998-4325		4455 Bird Ave
00006	Scott	818-231-2566	650-443-2211		543 Dale Ave

The table is lexicographically sorted on the row keys

Cells

```
hbase(main):009:0> create 'guru99', {NAME=>'e', VERSIONS=>6}
0 row(s) in 1.3790 seconds

=> Hbase::Table - guru99
hbase(main):010:0> put 'guru99', 'r1', 'e:c1', 'value', 10
0 row(s) in 0.0880 seconds

hbase(main):011:0> put 'guru99', 'r1', 'e:c1', 'value', 12
0 row(s) in 0.0090 seconds

hbase(main):012:0> put 'guru99', 'r1', 'e:c1', 'value', 14
0 row(s) in 0.0110 seconds

hbase(main):013:0> delete 'guru99', 'r1', 'e:c1', 11
0 row(s) in 0.0270 seconds
```

Creating Table guru99 and placing values in guru99

```
hbase(main):010:0> get 'guru99','r1', 'c1'
COLUMN
c1:
1 row(s) in 0.0440 seconds

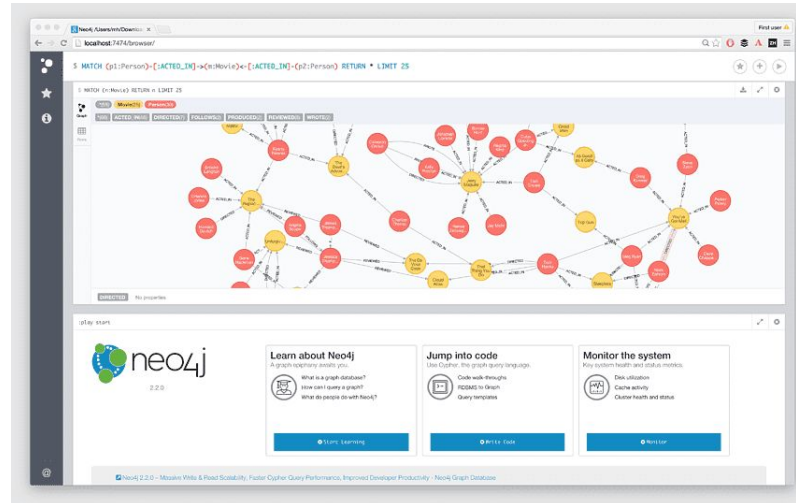
CELL
timestamp=30, value=value
```

getting records from 'guru99' using get

NoSQL Databases

Classification

- Graph databases :

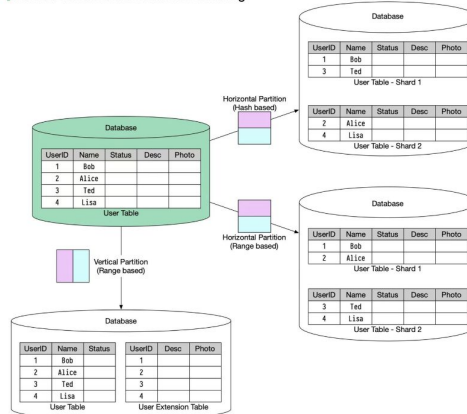


NoSQL Databases

Characteristics of databases

- Data partitioning:
 - Data divided into smaller, more manageable chunks (partitions/shards) based on a partitioning strategy.

Vertical & Horizontal Database Sharding



Key characteristics:

- Logical or physical separation of data based
- Each partition contains a subset of the complete dataset

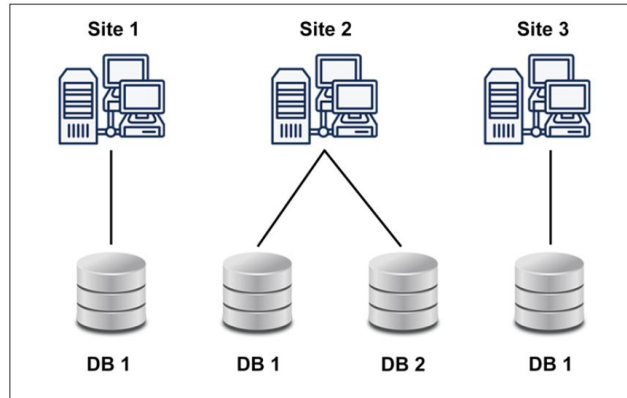
Purpose:

- Improved query performance
- More manageable data sizes
- Horizontal scalability

NoSQL Databases

Characteristics of databases

- **Data distribution**
 - Data is spread across multiple physical machines or nodes, typically in different locations.



Key characteristics:

- Geographic separation between nodes
- Replication of data across locations for availability/redundancy

Purpose:

- High availability
- Disaster recovery
- Geographic performance optimization



NoSQL Databases

Characteristics of databases

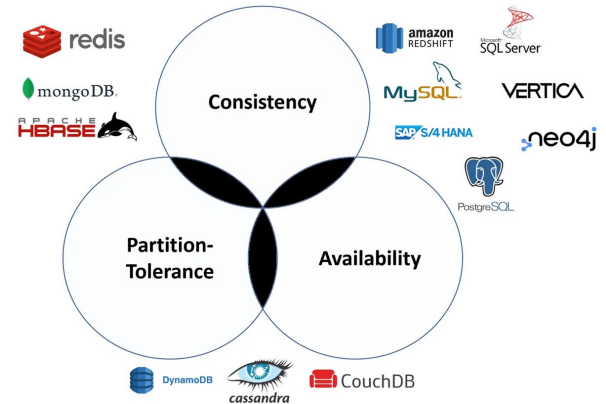
- **The CAP Theorem** => in any distributed system, it is not possible to satisfy these 3 properties at the same time :
 - **Consistency** : All servers see the same data at the same time (if you write something, everyone instantly sees it)
 - **Availability** : Every request gets a response, even if it's not the latest data (the system is always up and replies, even during problems)
 - **Partition tolerance** : The system still works even if parts of it can't communicate (like when a network cable is cut)
- **Rule** : We can only guarantee 2 out of the 3 at any moment, must sacrifice one if there's a failure (like a server crash or network problem)

NoSQL Databases

Characteristics of databases

- The CAP Theorem

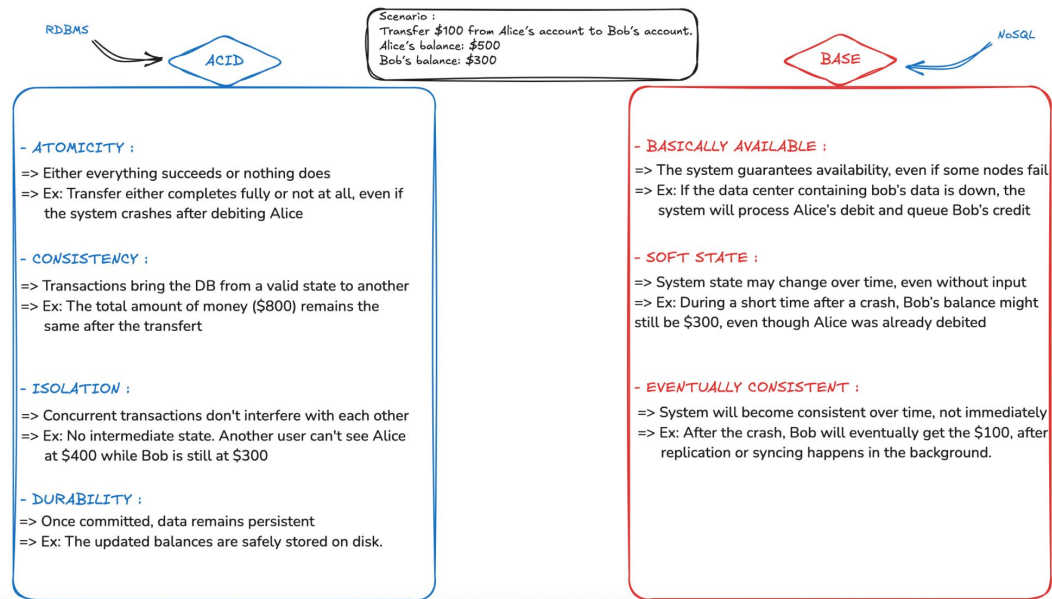
Choice	Example	Meaning
CP (Consistency + Partition)	HBASE, MongoDB	Always correct data, but might be unavailable during network issues
AP (Availability + Partition)	Amazon DynamoDB, Cassandra	Always responsive, but maybe outdated data sometimes
CA (Consistency + Availability)	Traditional databases (MySQL, PostgreSQL)	Needs perfect network – All clients can read and write on same data



NoSQL Databases

Characteristics of databases

- BASE Vs ACID





Part 3 :

Docker for Database
Environments



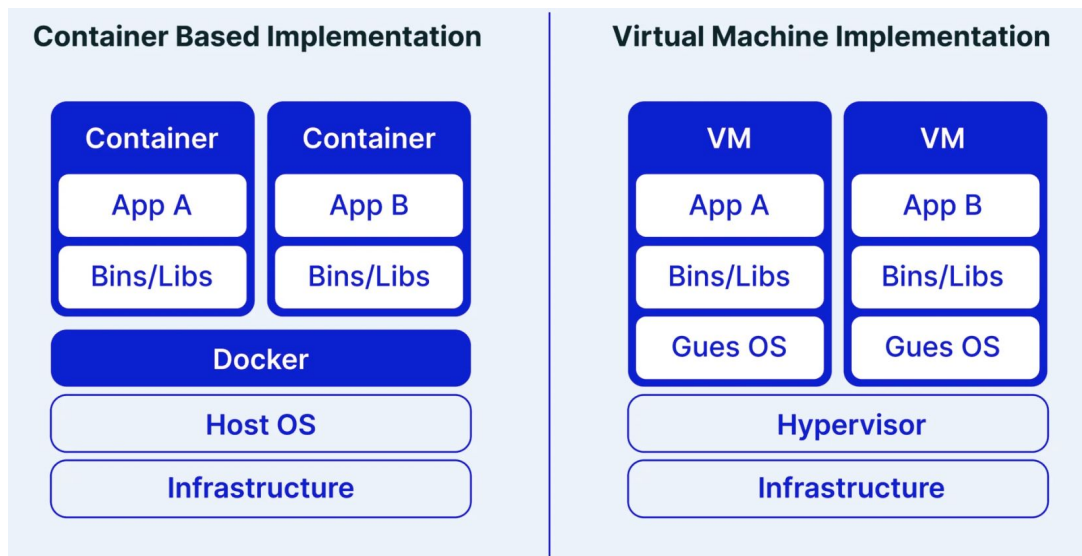
Docker for Database Environments

Introduction to containerization

- **Containerization** is a method of packaging and running applications along with their dependencies in a lightweight, portable, and isolated environment called a container
- Different from **virtualization** which is a process of creating an abstraction layer over hardware, allowing a single computer to be divided into multiple virtual computers.
- **Container** : standalone virtual machine, but without the overhead of a full Operating System. It includes:
 - The application (e.g., a database)
 - Its dependencies (libraries, binaries, config files)
 - Everything needed to run, regardless of the host system

Docker for Database Environments

Introduction to containerization





Docker for Database Environments

Containerization for databases

Why containerization in the context of databases ?

- Easy & Consistent Deployment, Portability
 - Containers work across different environments: Linux, Mac, Windows, cloud platforms.
 - The database will behave the same way on your machine, in staging, in production...
- Fast Testing and Development
 - Developers can test multiple versions of a database without complex installs.
 - Infrastructure as code: define your full DB setup in code and redeploy it anywhere.
 - You can isolate test environments easily.



Considerations for Production :

- Containers are **ephemeral**: by default, data is lost when the container is deleted.
- So for databases, it's critical to use volumes to persist data



Docker for Database Environments

Docker solution



What is Docker?

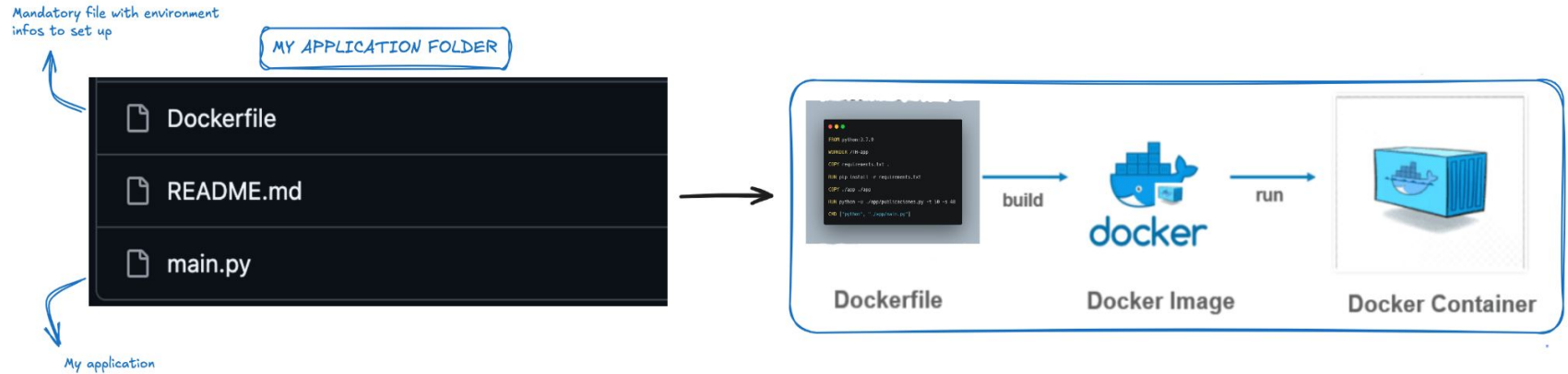
- An open-source platform that enables you to automate the deployment, scaling, and management of applications using lightweight & portable containers
- A Docker **container** bundles an application along with all its dependencies (libraries, configuration files, system tools) so it can run reliably across different computing environments

How Docker Works?

- A **Dockerfile** describes the environment and dependencies (possible to retrieve samples online)
- You build an image from that file
- You run that image as a container

Docker for Database Environments

Docker solution

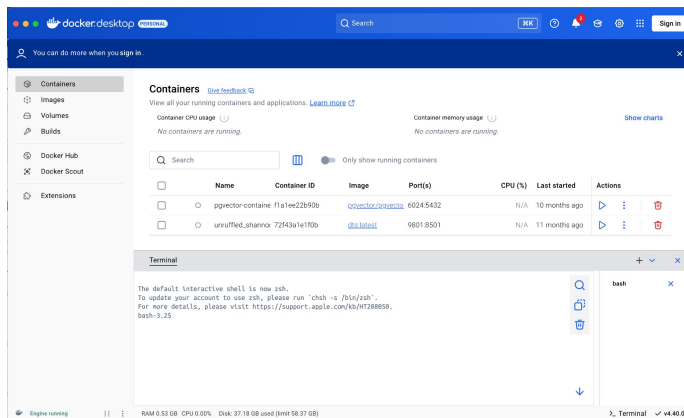


Docker for Database Environments

Installing docker

Desktop version :

- Windows : <https://docs.docker.com/desktop/setup/install/windows-install/>
- Mac OS : <https://docs.docker.com/desktop/setup/install/mac-install/>

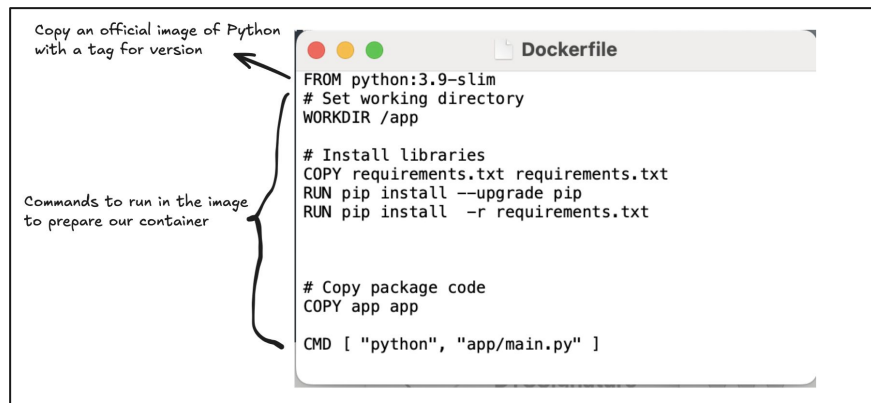


Docker for Database Environments

Building and running containers

Suppose we have a Python code (*main.py*) that we want to run inside a docker container.
We must complete **3 Steps** :

- 1- Create a **Dockerfile** in the same directory



Docker for Database Environments

Building and running containers

Suppose we have a Python code (*main.py*) that we want to run inside a docker container.
We must complete **2 Steps** :

- **2- Build** the Docker image

Open a terminal in the directory with your Dockerfile and run

```
bash
```

```
docker build -t my-python-app .
```

names the image

current directory as context

Docker for Database Environments

Building and running containers

Suppose we have a Python code (*main.py*) that we want to run inside a docker container. We must complete **3 Steps** :

- **3- Run** the Docker image in a container

```
bash
docker run -it --name my-container my-python-app
```

gives a name to the container

name of the image to run



NB : the container will stop running after completing if the app is only about running a script



Docker for Database Environments

Building and running containers

Other docker commands

- **List containers:**
 - `docker ps`
- **Check logs:**
 - `docker logs my-container`
- **Stop a container:**
 - `docker stop my-container`
- **Remove a container:**
 - `docker rm my-container`
- **Copy a remote image from Docker hub :**
 - `docker pull `THE_OFFICIAL_IMAGE_NAME``



Docker for Database Environments

Building and running containers

Exercise

- **Basic docker app :**
 - Build & run your own docker image that executes a python script. This project can inspire you : https://github.com/ababacaryoro/nosql_course/tree/dev/Docker-init
- **Docker app with Jupyter notebook:**
 - Create a docker app running a Jupyter notebook application on **port 9988** of your computer
 - Mount with your app a local folder containing your notebooks when running your application



Docker for Database Environments

Docker Compose : multi-container applications

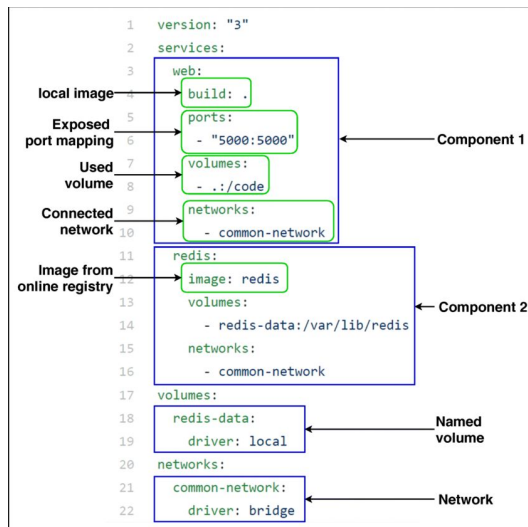
- When an application needs to run multiple services for different tasks => multiple docker applications
- Instead of running individual ***docker run*** commands for each service, Docker Compose lets you define them in a file called ***docker-compose.yml***
- Ex : a web app that needs a database, a cache, and maybe a background worker.
- Benefits :
 - Centralized configuration for all containers
 - Easy to bring up/down a multi-container setup
 - Networking between containers is automatic
 - Supports volume mounting, environment variables, port mappings

Docker for Database Environments

Docker Compose : multi-container applications

Steps to create a multi-container application

- 1- Create the *docker-compose.yml* file



Docker for Database Environments

Docker Compose : multi-container applications

Steps to create a multi-container application

- 2- Start the app
 - `docker-compose up` or `docker compose up` in MacOS
- Other commands :
 - Stop the App: `docker-compose down`
 - Build services from Dockerfiles : `docker-compose build`
 - View logs from services: `docker-compose logs`
 - List containers : `docker-compose ps`

Containers [Give feedback](#)

View all your running containers and applications. [Learn more](#)

Container CPU usage 0.01% / 1200% (12 CPUs available) Container memory usage 254.93MB / 7.48GB

Search

☐ Only show running containers

☐	Name	Container ID	Image	Port(s)
☐	pgvector-container	f1a1ee22890b	pgvector/pgvector:pg16	6024-5432
☒	multi-container	-	-	-
☐	db-1	5c66e6cb02c5	postgres:15	
☐	notebook-1	d717d4ae006a	jupyter/scipy-notebook	8888:8888



Docker for Database Environments

Docker Compose : multi-container applications

Exercise :

- Create a multi-container application running with a Jupyter notebook and a PostgreSQL app, with the following specifications :
 - Mount/create a volume for the folder of your notebooks with the Jupyter app
 - Create password & username with the PostgreSQL app (use a .env file)
 - Mount a persistent volume for the database
 - Run the container & open a new notebook with the Jupyter app
 - Using the notebook :
 - Connect to PostgreSQL and create a database “school”
 - Create 2 tables : students (columns name, birth date, class) and course (columns name, duration, level)
 - Add random data to theses tables
 - Check if was correctly added (load tables)



Part 4 :

Key-Value Stores with Redis



Key-Value Stores with Redis

Key-value store concepts

- A **Key-Value Store** is a simple type of NoSQL database where each piece of data is stored as a pair:
 - **Key** → Value
- **Structure :**
 - **Key:** Unique identifier (e.g., a string or number)
 - **Value:** Any kind of data (e.g., string, JSON, binary, image)
- **Example :**

```
"user:1001" → { "name": "Alice", "age": 29 }  
"session:xyz" → "abc123token"
```

Key-Value Stores with Redis

Key-value store concepts

Use cases





Key-Value Stores with Redis

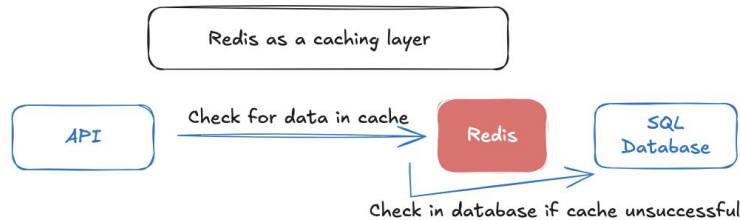
Key-value store concepts

- **Characteristics :**
 - **Very fast** due to $O(1)$ access
 - **Scalable** and simple to distribute across servers
 - **Easy to use** for developers
- **Limitations :**
 - No complex queries (e.g., JOINS)
 - Hard to model relationships
 - Not ideal for structured, relational data

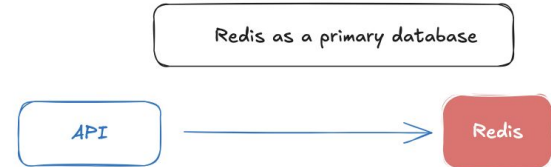
Key-Value Stores with Redis

Redis DB

- **Redis (Remote Dictionary Server)** : an open-source database management system designed to store, retrieve, and manipulate various key/value pair data
- Can be configured to store data on **disk**, however, it is primarily used for **in-memory** data management, making it much faster
- **Usage :**



Data is stored in memory => in case of crash, they will be lost in Redis but still available in the database

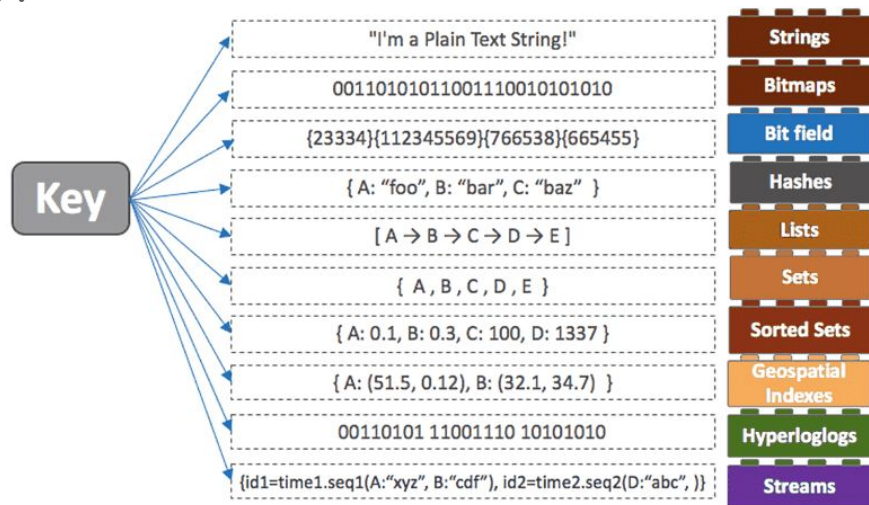


Data can be stored in disk => data is persisted and a system of replication is used to make data durable and available

Key-Value Stores with Redis

Redis DB

- Redis data types :





Key-Value Stores with Redis

Redis DB

- **Redis data organisation :**
 - No tables, collections or schema : just objects stores using keys/value system
- **Simulate common data organization :**
 - Use **data structures** and **keys naming conventions** to better manipulate data
- **Common Redis key naming conventions :**
 - Use namespacing with colon separators (:) -> 'user:123'; 'session:abc456'
 - Use lowercase letters and consistent casing : 'product:456' instead of 'Product:456'
 - Prefix by entity type or application : 'app1:user#1001'; 'app2:user#1001' (**colon** for namespaces, **sharp** for IDs)
 - Avoid spaces and special characters : Use underscores _ or colons ; instead.
- **TTL (Time To Live) :** defines how long a key should exist before it automatically expires and is deleted from the Redis database.
 - Can be defined when creating the (key, value)



Key-Value Stores with Redis

Hands-on Redis

- **Installing :**
 - Desktop client : <https://redis.io/open-source/>
 - Redis Cloud : <https://redis.io/cloud/>
 - Docker : https://redis.io/docs/latest/operate/oss_and_stack/install/install-stack/docker/
- **TODO :** install using docker compose
 - Install redis image => `image: redis:latest`
 - Add redis insight (GUI client) => `image: redis/redisinsight:latest`
 - Add Jupyter notebook



Key-Value Stores with Redis

Hands-on Redis

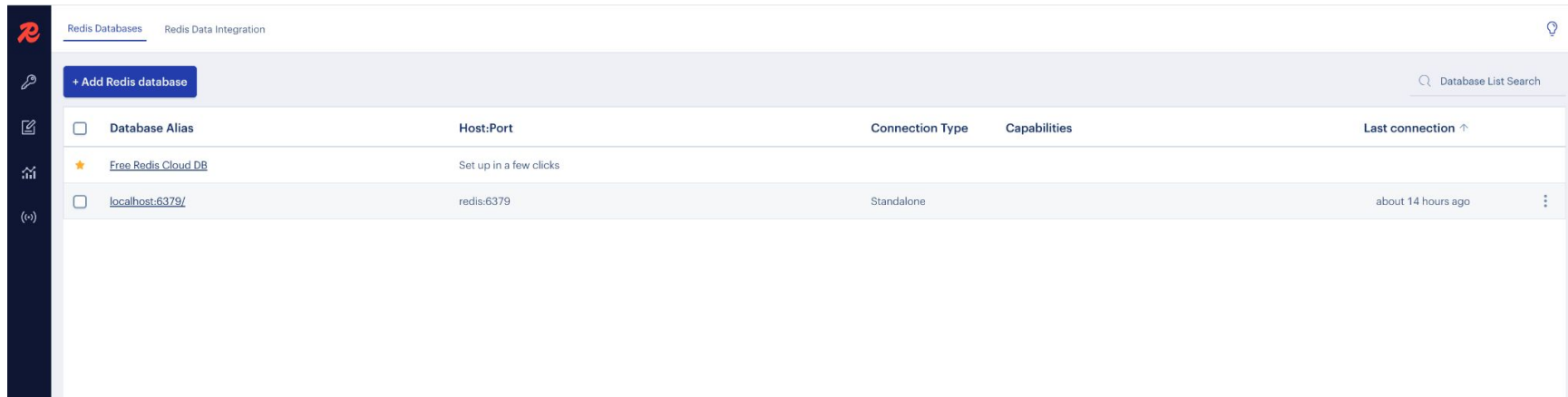
- **TODO** : install using docker compose

```
version: '3.9'
> Run All Services
services:
  > Run Service
  redis:
    image: redis:latest
    container_name: redis-server
    ports:
      - "6379:6379"
    volumes:
      - redis-data:/data
  > Run Service
  redisinsight:
    image: redis/redisinsight:latest
    container_name: redis-insight
    ports:
      - "5540:5540"
    depends_on:
      - redis
  > Run Service
  notebook:
    image: jupyter/scipy-notebook
    ports:
      - "9988:8888"
    volumes:
      - ./notebooks:/home/jovyan/work
    environment:
      - JUPYTER_ENABLE_LAB=yes
volumes:
  redis-data:
```

Key-Value Stores with Redis

Hands-on Redis

- Redis insight GUI client



The screenshot displays the Redis Insight web interface. On the left is a dark sidebar with navigation icons. The main content area has a header with 'Redis Databases' and 'Redis Data Integration' tabs. Below the header is a '+ Add Redis database' button and a 'Database List Search' input field. A table lists the configured Redis databases with columns for 'Database Alias', 'Host:Port', 'Connection Type', 'Capabilities', and 'Last connection'. Two databases are listed: 'Free Redis Cloud DB' and 'localhost:6379/'.

Database Alias	Host:Port	Connection Type	Capabilities	Last connection ↑
★ Free Redis Cloud DB	Set up in a few clicks			
☐ localhost:6379/	redis:6379	Standalone		about 14 hours ago

Key-Value Stores with Redis

Hands-on Redis

- Redis insight GUI client

Tab to visualize data

Tab for queries

Databases / localhost:6379/ db0

0.52% 0 2 MB 8 2

1

Tutorials: Intro to search Basic use cases Intro to vector search

Raw mode Group results Run

Clear Results

HSET person:1 name Alice	02-02-17 4 May 2025	19.537 msec	🗑️ ▶️ ▼
HGET user:112	02-01-41 4 May 2025	3.826 msec	🗑️ ▶️ ▼
GET user:112	02-01-25 4 May 2025	4.236 msec	🗑️ ▶️ ▼
HSET user:112 name Alice age 3	02-01-01 4 May 2025	1.963 msec	🗑️ ▶️ ▼

Key-Value Stores with Redis

Hands-on Redis

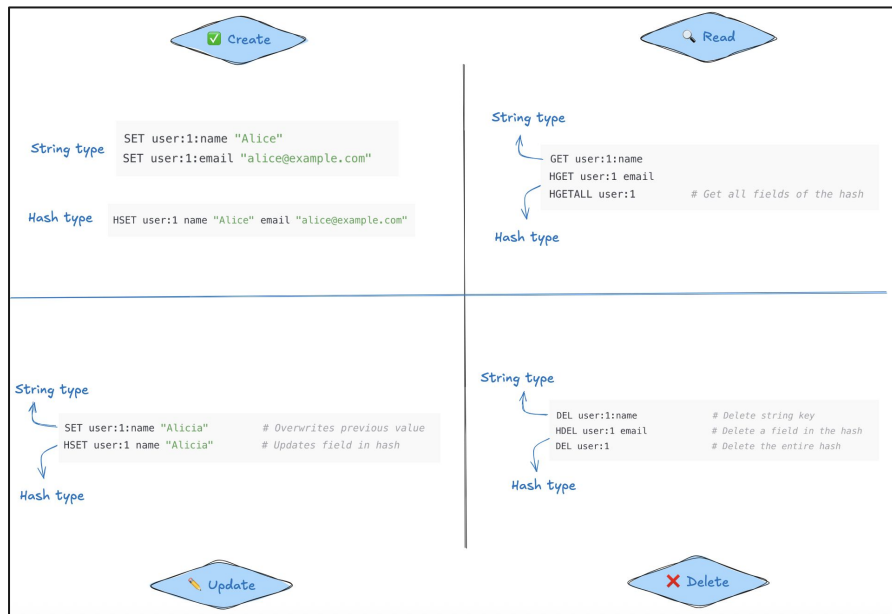
- **Basic commands :** Each data structure in Redis has its own unique set of operations that can be performed on it :

Category	Command	Description	Example
🔑 Keys	SET key value	Store a value	SET user:1 "Alice"
	GET key	Retrieve a value	GET user:1
	DEL key	Delete a key	DEL user:1
	EXISTS key	Check if key exists	EXISTS user:1
	EXPIRE key seconds	Set TTL	EXPIRE user:1 60
	TTL key	Get remaining TTL	TTL user:1
📄 Strings	APPEND key value	Add to string	APPEND user:1 " Smith"
🔢 Incr	INCR key / DECR key	Increment/Decrement	INCR counter
🗑 Hashes	HSET key field value	Set hash field	HSET user:1 name Alice
	HGET key field	Get hash field	HGET user:1 name
📋 Lists	LPU SH key value	Add to list start	LPU SH todos "buy milk"
	LRANGE key 0 -1	Get all list items	LRANGE todos 0 -1
🔢 Sets	SADD key value	Add to set	SADD tags redis
	SMEMBERS key	Get set members	SMEMBERS tags

Key-Value Stores with Redis

Hands-on Redis

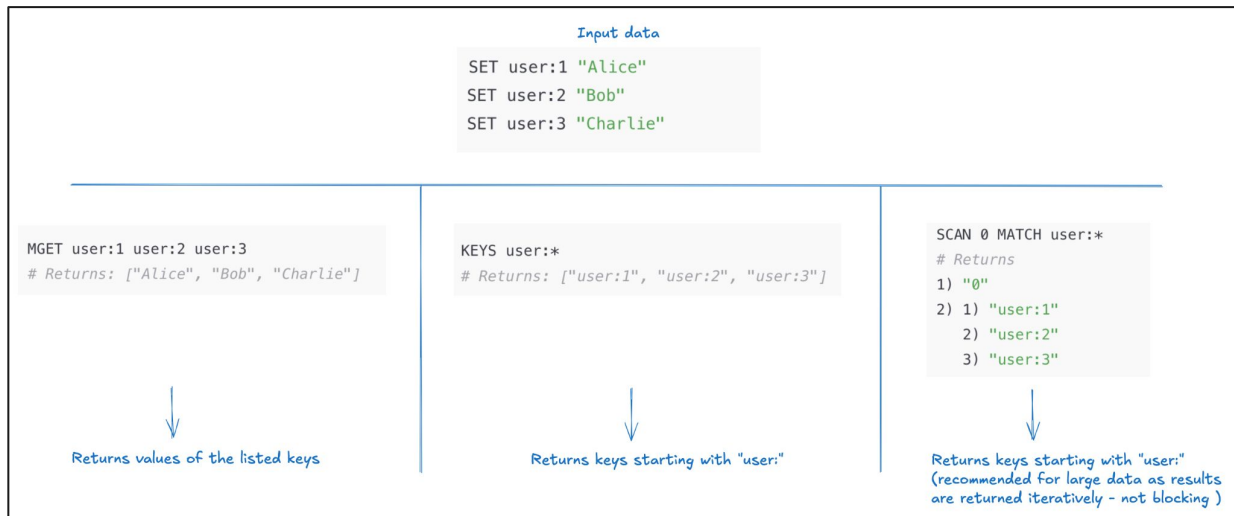
- CRUD (Create, Read, Update, Delete):



Key-Value Stores with Redis

Hands-on Redis

- Retrieve multiple keys :



Key-Value Stores with Redis

Hands-on Redis

- **Basic commands**: using Python client



```
!pip install redis
import redis
r = redis.Redis(host='redis', port=6379, db=0)
```

Create

```
# Set a key
r.set("user:1", "Alice")

# Set multiple keys
r.mset({"user:2": "Bob", "user:3": "Charlie"})
```

Read

```
# Get a value
r.get("user:1") # b'Alice'

# Get multiple values
r.mget(["user:1", "user:2"]) # [b'Alice', b'Bob']

# List all matching keys
r.keys("user:*") # [b'user:1', b'user:2', b'user:3']
```

Update

```
# Overwrite value
r.set("user:1", "Alice Smith")

# Increment value (e.g., score)
r.set("score", 100)
r.incr("score") # 101
r.incrby("score", 5) # 106
```

Delete

```
# Delete a single key
r.delete("user:1")

# Delete multiple keys
r.delete("user:2", "user:3")
```



Key-Value Stores with Redis

Hands-on Redis

- **Application :**
 - Instructions : https://github.com/ababacaryoro/nosql_course/blob/dev/Redis/redis_practical_exercise.md
 - Actions to do :
 - Load data into Redis using Docker
 - Interact with Redis using Python
 - Write queries to analyze data



Part 5 :

Document Stores with
MongoDB



Document Stores with MongoDB

Document Stores concepts

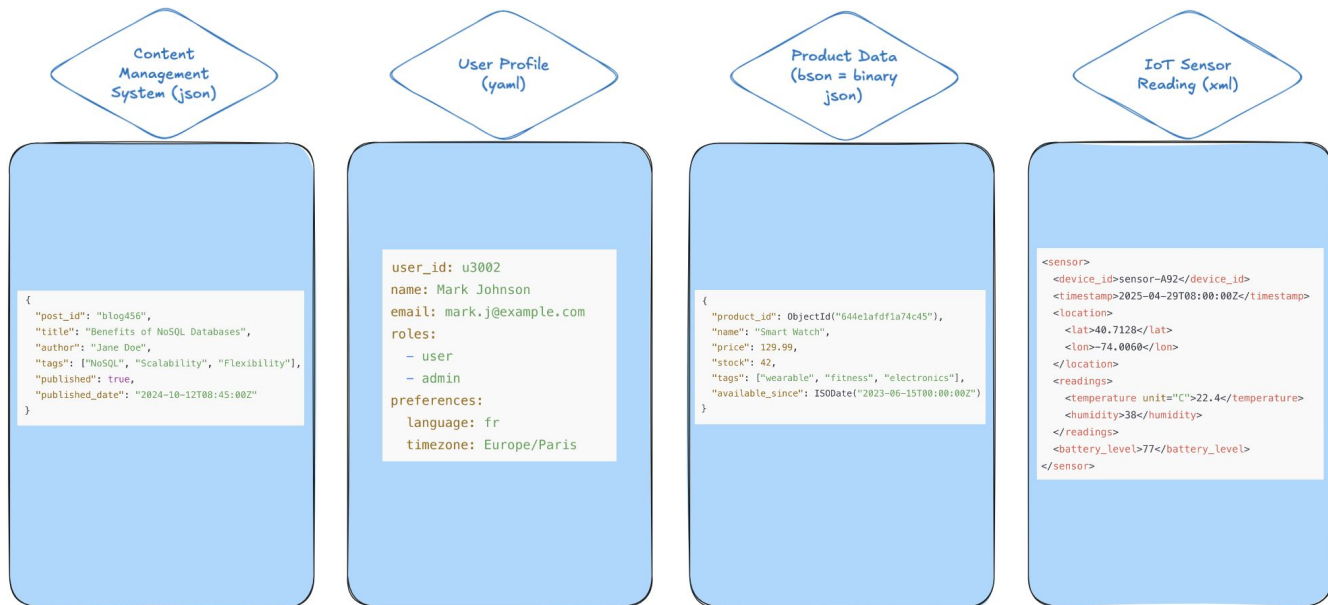
- A **Document Store** is a type of NoSQL database designed to store, retrieve, and manage semi-structured data in the form of documents, typically in JSON, BSON, or XML format.
- Each **document** is :
 - A **self-contained unit** (like a row in SQL)
 - Can have **nested fields**, that can be **scalar** (string, number), **array**, or **nested objects**
 - Documents are grouped into **collections** (equivalent to tables in SQL)
- **Example of document :**

```
{
  "id": "user123",
  "name": "Alice",
  "email": "alice@example.com",
  "roles": ["admin", "editor"],
  "profile": {
    "age": 29,
    "location": "Paris"
  }
}
```

Document Stores with MongoDB

Document Stores concepts

Use cases





Document Stores with MongoDB

Document Stores concepts

- **Characteristics :**
 - **Schema-less:** documents can vary in structure.
 - **Rich query language:** supports filtering, projection, and aggregation
 - **Indexing:** you can index fields inside documents for fast queries
 - **Scalability:** designed for horizontal scaling (via sharding)
- **Limitations :**
 - **No strong schema enforcement :** documents can have inconsistent structures within 1 same collection
 - **Data duplication & redundancy :** data is often denormalized (copied across documents)
 - **Join limitations :** most document databases do not support joins natively like SQL



Document Stores with MongoDB

MongoDB

- **MongoDB**: a popular open-source, document-oriented NoSQL database
- **Key features** :
 - Stores data as **BSON** (binary JSON)
 - **Schema-less**: flexible document structure
 - Ideal for **semi-structured** or **evolving** data models



Document Stores with MongoDB

MongoDB

- **Data structure :**
 - **Database** → contains multiple **collections**
 - **Collection** → contains multiple **documents**
 - **Document** → a **JSON-like object** (`{ "name": "Alice", "age": 30 }`)
- **Common data types :**

Data Type	Example	Notes
String	<code>"name": "Alice"</code>	Most commonly used data type
Integer	<code>"age": 30</code>	32-bit or 64-bit depending on the server
Double	<code>"price": 19.99</code>	For floating-point numbers
Boolean	<code>"active": true</code>	<code>true</code> or <code>false</code>
Date	<code>"created_at": ISODate(...)</code>	Stores dates with millisecond precision
ObjectId	<code>"_id": ObjectId(...)</code>	Unique identifier for each document
Array	<code>"tags": ["tech", "news"]</code>	List of values
Embedded Document	<code>"address": { "city": "Paris" }</code>	Document inside a document
Null	<code>"deleted_at": null</code>	Explicitly represents a missing field

Document Stores with MongoDB

MongoDB

- Data modeling options :

EMBEDDING

"Orders" collection

```
{
  "_id": ObjectId("..."),
  "order_id": "ORD123",
  "total": 89.99,
  "customer": {
    "customer_id": "CUST45",
    "name": "Alice",
    "email": "alice@example.com"
  }
}
```

REREFENCING

"Orders" collection

```
{
  "_id": ObjectId("..."),
  "order_id": "ORD123",
  "total": 89.99,
  "customer_id": "CUST45"
}
```

"Customers" collection

```
{
  "_id": "CUST45",
  "name": "Alice",
  "email": "alice@example.com"
}
```

🤔 Decision Guide

Criteria	Embed	Reference
Data is always read together	✅ Yes	❌ No (requires join or extra query)
Data changes frequently	❌ No (duplication risk)	✅ Yes (update once)
One-to-few relationship	✅ Embed is preferred	❌ Reference adds overhead
One-to-many or many-to-many	❌ Embedding can bloat document	✅ Use referencing



Document Stores with MongoDB

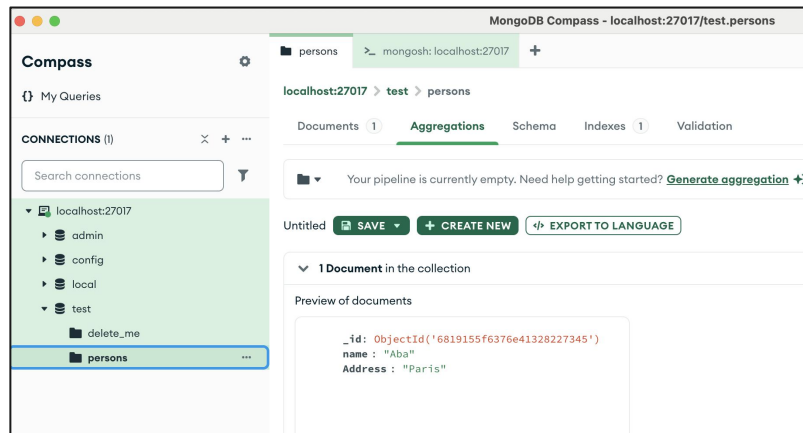
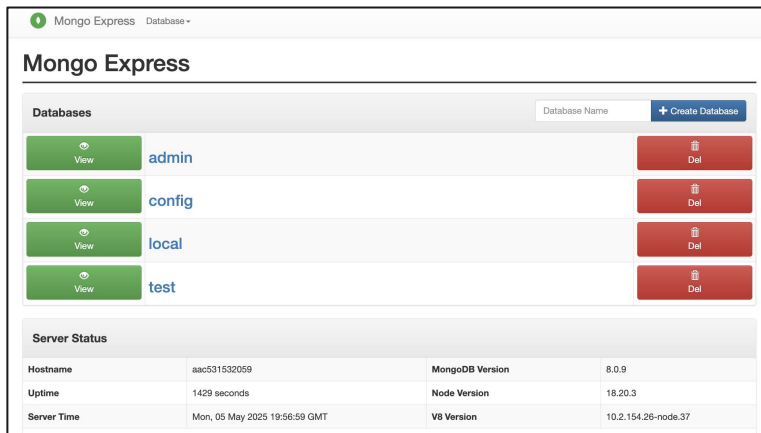
Hands-on MongoDB

- **Installing :**
 - Desktop client : <https://www.mongodb.com/docs/manual/installation/>
 - Docker :
 - https://hub.docker.com/_/mongo
 - <https://www.mongodb.com/docs/manual/tutorial/install-mongodb-community-with-docker/>
- **TODO : install using docker compose**
 - Install MongoDB image => `image: mongo:latest`
 - Add MongoDB (GUI client) => `image: mongo-express:latest` or install Mongo Compass (<https://www.mongodb.com/try/download/compass>)
 - Add Jupyter notebook

Document Stores with MongoDB

Hands-on MongoDB

- MongoDB GUI clients (MongoDB Express & MongoDB Compass)



Document Stores with MongoDB

Hands-on MongoDB

- **Basic commands :** MongoDB shell (***mongosh***) => on MongoDB Compass

The image shows two side-by-side screenshots of MongoDB tools. The left screenshot is from MongoDB Compass, displaying the 'test' database and the 'customers' collection. The right screenshot is from the MongoDB Shell (mongosh), showing the same database and collection.

MongoDB Compass - localhost:27017/test

test +

localhost:27017 > test

Sort by Collection Name

Open MongoDB shell Create collection View

customers

Storage size:	Documents:	Avg. document size:	Indexes:	Total index size:
16.24 MB	25 K	1.66 kB	1	266.24 kB

MongoDB Shell (mongosh)

test >_ mongosh: localhost:27017 +

```
>_MONGOSH
> use test
< already on db test
test> |
```



Document Stores with MongoDB

Hands-on MongoDB

- **Basic commands :** MongoDB shell (*mongosh*) => on MongoDB Compass

Database commands

```
// Show all databases
show dbs

// Switch to or create a new database
use myDatabase

// Show current database
db
```

Collection commands

```
// Show collections in the current database
show collections

// Create a new collection
db.createCollection("myCollection")

// Drop (delete) a collection
db.myCollection.drop()
```

Document Stores with MongoDB

Hands-on MongoDB

- Basic commands : MongoDB shell (*mongosh*) : CRUD



```
// Insert one document
db.users.insertOne({ name: "Alice", age: 25 })

// Insert multiple documents
db.users.insertMany([
  { name: "Bob", age: 30 },
  { name: "Charlie", age: 35 }
])
```



```
// Find all documents
db.users.find()

// Find with filter
db.users.find({ age: { $gt: 30 } })

// Find one document
db.users.findOne({ name: "Alice" })
```

Document Stores with MongoDB

Hands-on MongoDB

- Basic commands : MongoDB shell (*mongosh*) : CRUD

```
// Update one document
db.users.updateOne({ name: "Alice" }, { $set: { age: 26 } })

// Update multiple documents
db.users.updateMany({}, { $inc: { age: 1 } }) // Increment age for all users
```



```
// Delete one document
db.users.deleteOne({ name: "Bob" })

// Delete multiple documents
db.users.deleteMany({ age: { $lt: 30 } })
```





Document Stores with MongoDB

Hands-on MongoDB

- **Basic commands : MongoDB shell (*mongosh*) : Filters**
 - Can be used inside the `.find()` method

Operator	Description	Example
<code>\$eq</code>	Equal to	<code>{ age: { \$eq: 25 } }</code>
<code>\$ne</code>	Not equal to	<code>{ age: { \$ne: 25 } }</code>
<code>\$gt</code>	Greater than	<code>{ age: { \$gt: 18 } }</code>
<code>\$gte</code>	Greater than or equal to	<code>{ age: { \$gte: 18 } }</code>
<code>\$lt</code>	Less than	<code>{ age: { \$lt: 30 } }</code>
<code>\$lte</code>	Less than or equal to	<code>{ age: { \$lte: 30 } }</code>
<code>\$in</code>	In array of values	<code>{ age: { \$in: [18, 25, 30] } }</code>
<code>\$nin</code>	Not in array of values	<code>{ age: { \$nin: [18, 25, 30] } }</code>



Document Stores with MongoDB

Hands-on MongoDB

- **Basic commands : MongoDB shell (*mongosh*) : Filters**
 - Can be used inside the `.find()` method

Operator	Description	Example
<code>\$eq</code>	Equal to	<code>{ age: { \$eq: 25 } }</code>
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<code>\$gt</code>	Greater than	<code>{ age: { \$gt: 18 } }</code>
<code>\$gte</code>	Greater than or equal to	<code>{ age: { \$gte: 18 } }</code>
<code>\$lt</code>	Less than	<code>{ age: { \$lt: 30 } }</code>
<code>\$lte</code>	Less than or equal to	<code>{ age: { \$lte: 30 } }</code>
<code>\$in</code>	In array of values	<code>{ age: { \$in: [18, 25, 30] } }</code>
<code>\$nin</code>	Not in array of values	<code>{ age: { \$nin: [18, 25, 30] } }</code>



Document Stores with MongoDB

Hands-on MongoDB

- **Basic commands : MongoDB shell (*mongosh*) : Filters**
 - Can be used inside the `.find()` method

Operator	Description	Example
<code>\$and</code>	Logical AND	<code>{ \$and: [{age: {\$gt:18}}, {age: {\$lt:30}}] }</code>
<code>\$or</code>	Logical OR	<code>{ \$or: [{age: {\$lt:18}}, {age: {\$gt:65}}] }</code>
<code>\$not</code>	Negate a condition	<code>{ age: { \$not: { \$gt: 18 } } }</code>
<code>\$nor</code>	None of the conditions are true	<code>{ \$nor: [{age: {\$gt:18}}, {name: "Alice"}] }</code>

Document Stores with MongoDB

Hands-on MongoDB

- Basic commands : using Python client

Install client and connect

```
!pip install pymongo dotenv
from pymongo import MongoClient
import os
from dotenv import load_dotenv, find_dotenv

load_dotenv(find_dotenv())

usr = "root" # os.environ.get("MONGO_INITDB_ROOT_USERNAME")
pwd = "pwd" # os.environ.get("MONGO_INITDB_ROOT_PASSWORD")
url = f"mongodb://{usr}:{pwd}@mongo:27017"
client = MongoClient(url)
```

Create database & collection

```
db = client["school"]
students = db["students"]
```

Insert document

```
new_student = {
    "name": "Alice",
    "age": 22,
    "class": "Math",
    "grades": [15, 17, 19]
}
insert_result = students.insert_one(new_student)
```

Read documents

```
# Get all students
all_students = students.find()
for student in all_students:
    print(student)

# Find one student
student = students.find_one({"name": "Alice"})
print("Found:", student)
```

Update

```
students.update_one(
    {"name": "Alice"},
    {"$set": {"age": 23}}
)

# Verify update
updated_student = students.find_one({"name": "Alice"})
print("Updated:", updated_student)
```

Delete

```
students.delete_one({"name": "Alice"})
print("Deleted Alice")
```



Document Stores with MongoDB

Hands-on MongoDB

- **Application 1 : MongoDB install & CRUP operations**
 - Instructions :
https://github.com/ababacaryoro/nosql_course/blob/dev/MongoDB/mongodb_practical_exercise1.md
 - Actions to do :
 - Extract data, transform and load on MongoDB using Docker
 - Interact with data using Mongo shell
 - Package & Deploy your code